Required: BottomUpFilling ⊑ Filling, 1.0 mL; the barrel 6.35 mm wide, its bottom 11 mm above the nest

pre: empty | post: content 1.0 mL | keep: −2 ≤ immersion ≤ 3

Actions from the Capability Description

Dosing: content += V at a rate r, 0.1 to 5 mL/s; immersion = y − (70 − 11) + 31.6 · content

Positioning: the needle to y at 12.5 mm/s; y ≤ 70 − 11 − 1

Goal regression: content = 1.0 mL

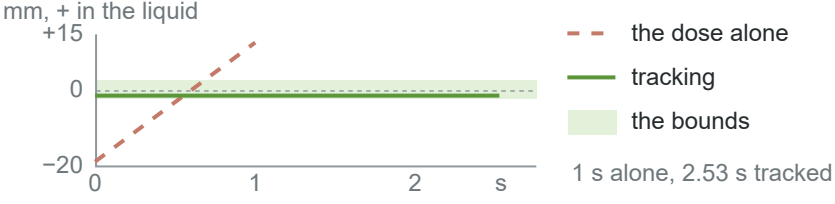
Dosing alone at 40 mm: immersion −19 → +12.6

breaks the bound at its start and end: not applied

Tracking: Dosing ‖ Positioning

- 31.6 · r = 12.5, so r = 0.396 mL/s
- starts at y_0 in [57, 58]: 57.5
- rises 31.6 mm in 2.53 s, to 25.9

In time: how deep the needle is in the liquid



The planned skill, checked in time, *Described*

MoveAxis 57.5 → Dispense 1.0 at 0.396 mL/s; MoveAxis 25.9 → Home